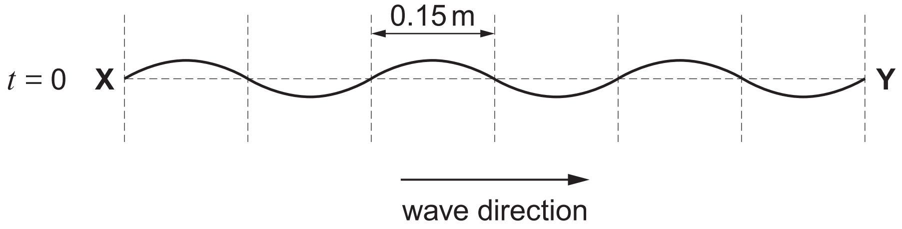
Examiner only

4. (a) A progressive wave is travelling from left to right on a stretched string. A snapshot of the portion **XY** of the string at time $t = 0$ is shown.



(i) State the value of the wavelength. [1]

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(ii) The periodic time is 100 ms. Calculate the **speed** of the waves. [2]

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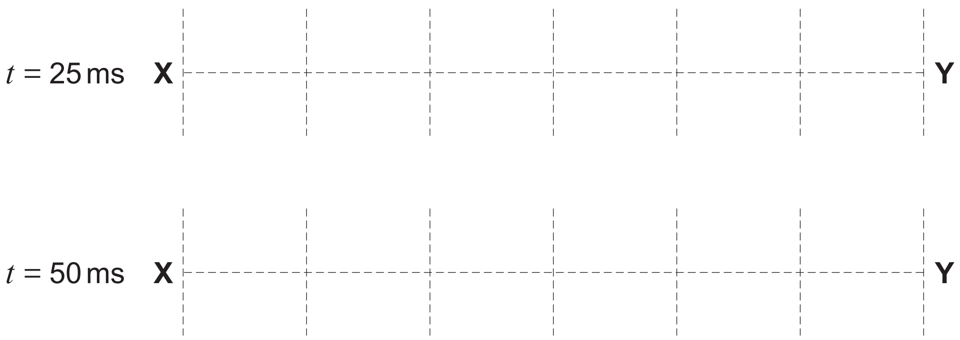
(iii) Carefully sketch, on the grids below, the wave along **XY** at times $t = 25$ ms and $t = 50$ ms. [2]



(b) If the string is attached to a fixed point to the right of **Y**, a stationary wave is observed on **XY**.

(i) Explain how this stationary wave arises. [2]

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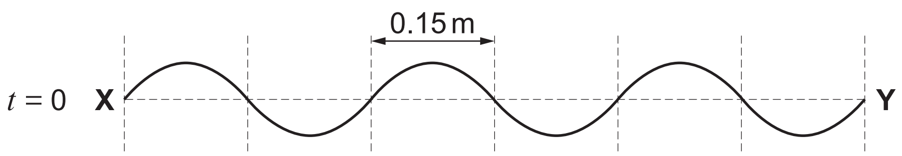
(ii) The diagram below shows **XY** at an instant (t = 0) when the stationary wave has maximum amplitude.



Carefully sketch, on the grids below, the wave along **XY** at times t = 25 ms and t = 50 ms. [2]